Customer Shopping Behaviour Analysis

Project Report

1. Project Report

In this project, I analysed customer shopping behaviour using a dataset containing 3,900 purchase records across different product categories. My aim was to understand how customers shop, what they prefer to buy, how much they spend, and whether subscription status is related to their purchasing habits. The analysis was intended to turn the available data into useful insights that can support business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Missing Data: 37 values in Review Rating column
- Key Features:
    - Customer Demographics (Age, Gender, Location, Subscription Status)
    - Purchase Details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
    - Shopping Behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

3. Exploratory Data Analysis using Python

I began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using *pandas*.
- **Initial Exploration:** Use *df.info ( )* to check structure and *.describe ( )* for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the *Review Rating* column using the median rating of each product category.
- **Column Standardization:** Renamed columns to 'snake-case' for better readability & documentation.
- **Feature Engineering:**
    - Created *age_group* column by binning customer ages.
    - Created *purchase_frequency_days* columns from purchase data.
- **Data Consistency Check:** Verified if *discount applied* & *promo_code_used* were redundant; dropped *promo_code_used.*

4. Data Analysis using SQL (Business Transactions)

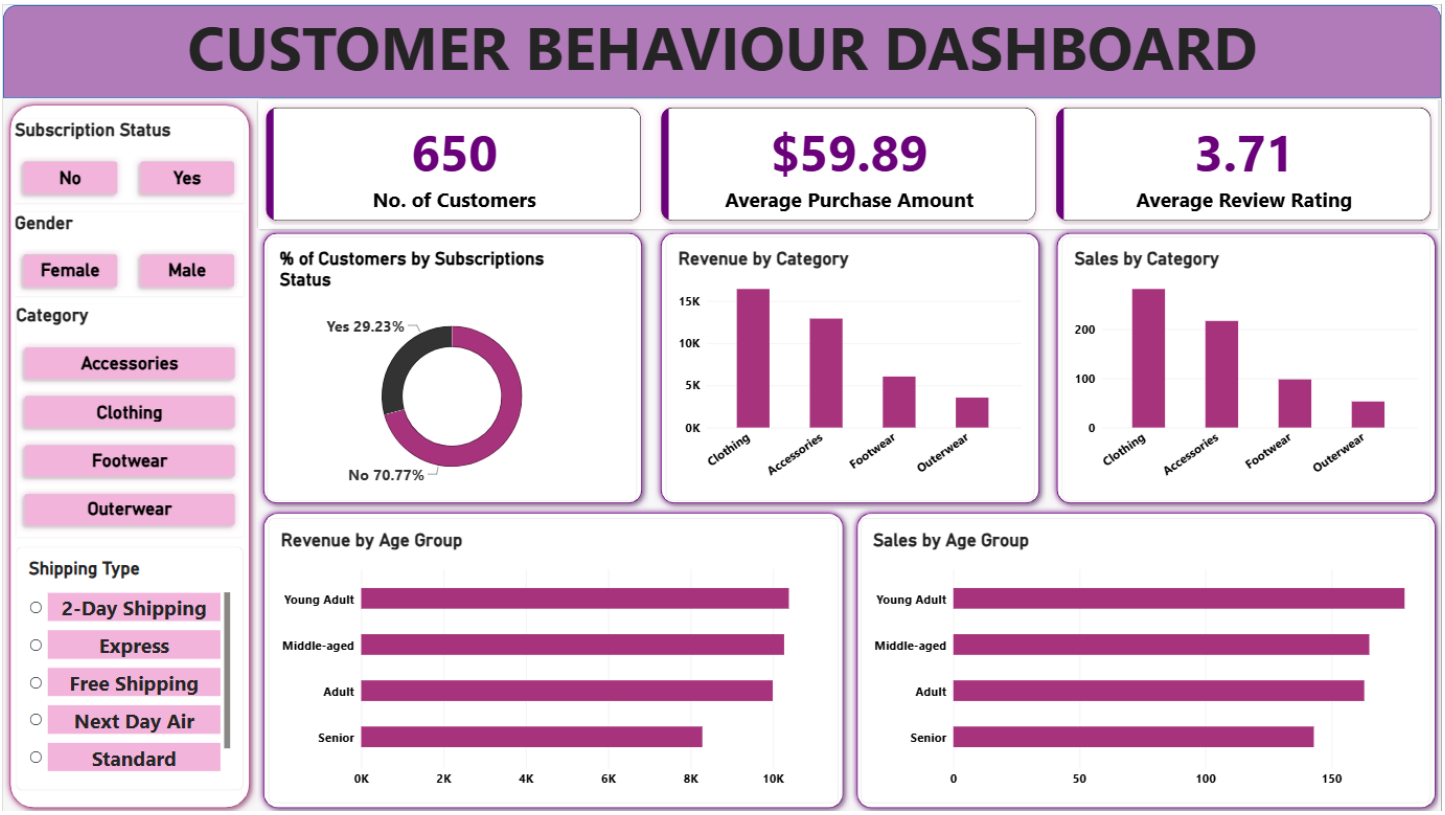
After preparing the data, I used MySQL to explore business questions about customer spending, product performance, loyalty, discounts, and subscriptions. The main queries covered the following areas:

- **Revenue by Gender:** Compared total revenue generated by male vs female customers.
- **High-Spending Discount Users:** Identified customers who used discounts but still spent above the average purchase amount.
- **Top 5 products by Rating:** Found products with the highest average review ratings.
- **Shipping Type Comparison:** Compared average purchase amounts between Standard & Express shipping.
- **Subscribers vs Non-Subscribers:** Compared average spend & total revenue across subscription status.
- **Discount Dependent Products:** Identified 5 products with the highest percentage of discounted purchases.

- **Customer Segmentation:** Classified customers into New, Returning & Loyal segments based on purchase history.
- **Top 3 products per Category:** Listed the most purchased products within each category.
- **Repeat Buyers & Subscriptions:** Checked whether customers with >5 purchases are more likely to subscribe.
- **Revenue by Age Group:** Calculated total revenue contribution of each age group.

5. Dashboard in Power BI

I used Power BI to bring the analysis together in an interactive dashboard. The dashboard provides a visual way to explore customer shopping patterns and compare important areas such as spending, product preferences, discounts, and subscriptions.



6. Business Recommendations

- **Boost Subscriptions:** Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs:** Reward repeated buyers to move them into “Loyal Segment”
- **Review Discount Policy:** Balance sales boosts with margin control.
- **Product Positioning:** Highlight top rated & best-selling products in campaigns.
- **Targeted Marketing:** Focus efforts on high revenue age groups & express shipping users.